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Classification and Concentration Prediction of VOC Gases Based on Sensor Array with Machine Learning Algorithms

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Abstract— **In this work, a facile method is proposed to monitor the freshness of meat and fruits by combining a gas sensor array with machine learning algorithms, where the sensor array consists of four commercial metal oxide-based gas sensors. Back-propagation neural network (BPNN) is used for gas classification, and the average accuracy can reach 98.8%. To obtain more effective prediction of VOC gas concentration (ethanol, trimethylamine, and ammonia), four algorithms including BPNN, radial basis function neural network (RBFNN), support vector machine (SVM), and hybrid LDA-SVM, which is a combination of SVM and linear discriminant analysis (LDA) are implemented, which are trained with the same training set. By analyzing and comparing the prediction results of these four algorithm models, the RBFNN achieves the peak performance for the concentration predictions of ethanol and ammonia, and the average relative errors are less than 5% and 6.5%, respectively. For trimethylamine (TMA) concentration prediction, the average relative error of RBFNN is equal to 4.41%, which is better than 5.11% of SVM, while the mean absolute error of RBFNN is slightly inferior to SVM. Therefore, the classification accuracy of the gas type by BPNN and the prediction accuracy of gas concentration by RBFNN can meet the requirement of distinguishing the freshness of food.**

I. Introduction

Food freshness has an important impact on daily diet safety and human health. Meat and vegetables are rich in proteins, fats, sugars, carbohydrates, organic acids, etc., which have high water activity and are easily contaminated by environmental microorganisms, leading to food corruption. Meat and fish spoilage will release a variety of volatile organic components, such as ammonia and trimethylamine (TMA), so we could find correlation with meat spoilage by using multi-sensors; and gas sensors can also measure low concentrations of ethanol, methanol, etc. to judge the maturity of fruits and vegetables [1].

Metal oxide gas sensors have many advantages, such as simple production, fast response, high sensitivity, etc., and can meet the requirements of identifying target gases related to food freshness [2-4]. Gas sensor arrays based on nanomaterials can overcome the shortcomings of poor selectivity of a single device and can improve the accuracy of gas detection [5-7].

In recent years, the combination of gas sensor arrays and machine learning algorithms to make electronic noses (EN) has attracted a lot of attention and has been applied in some fields. Back-propagation neural network (BPNN) is one of the basic neural networks, which has been widely used for gas classification and concentration prediction. It was proposed as early as 1970, but did not receive much attention until a joint paper was published by David Rumelhart, et.al. in 1986 [8]. BP networks can learn and store a large number of input-output mode mappings without the need to reveal mathematical equations describing such mappings in advance. J. Moody and C. Darken proposed another neural network algorithm based on radial basis function in 1988, which is radial basis function neural network (RBFNN) [9]. RBFNN is a typical local approximation network and can approximate any non-linear function with arbitrary precision like BPNN. However, Poggio and Girosi have proven that RBF networks have existence and uniqueness of best approximation, while BP networks do not have the best approximation property [10].

Support Vector Machine (SVM) is a type of generalized linear classifier, which does not belong to the category of neural networks. In order to achieve data estimation, it is to find a regression plane so that all the data of a set is closest to the plane [11]. As an alternative method to NN, the SVM method developed by Vapnik [12], defined the ε-insensitive loss function and obtained the ε-SVR algorithm, which can provide an effective method to improve generalization performance of neural networks. Linear discriminant analysis (LDA) is a classic linear learning method. It was first proposed by Fisher in 1936 on the problem of binary classification, also known as Fisher linear discriminant [13]. LDA can perform supervised dimensionality reduction on the data and choose the projection direction with the best classification performance.

In this work, based on reading a large number of references, four metal oxide gas sensors were used to form a sensor array, and four machine algorithms, BPNN, RBFNN, SVM, and LDA were applied, in order to identify ammonia, TMA and ethanol and predict their concentrations.

II. Method and System

In this work, based on cross-reactivity and sensitivity to three target gases: ethanol, ammonia and TMA, four commercial gas sensors based on metal oxides (such as SnO_2), MQ-3B, MQ136, MQ137, and MQ-4B were selected to form the sensor array — all four types of sensors were purchased from China Zhengzhou Weisheng Electronic Technology Co.,

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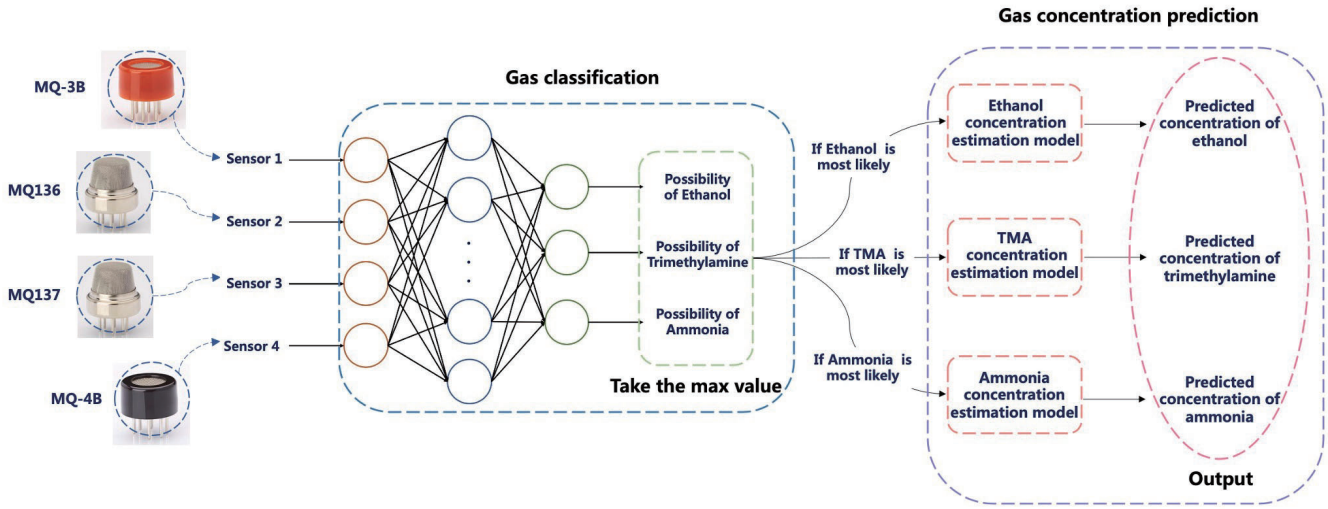


Figure 1. The structure of the system in this work.

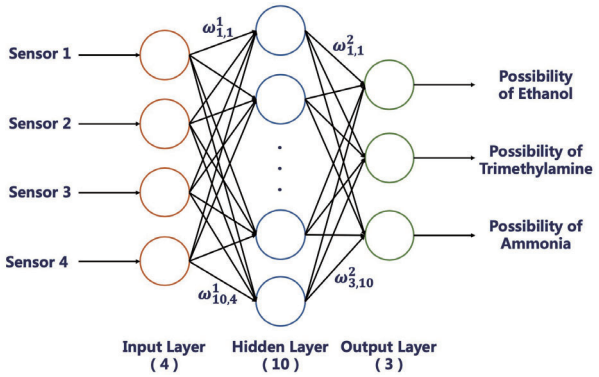


Figure 2. Topological structure of BPNN.

Ltd. By collecting the responses of the gas sensor array for the target gases and importing the data into the trained models, the identification and concentration prediction of the target gases can be completed.

Figure 1 shows the structure of the overall system in this work. The basic idea is to first input the collected responses of the gas sensor array to the BP neural network for gas identification, and then import the responses into the corresponding gas concentration prediction model based on the identification result to estimate its concentration, final output the gas type and its concentration.

A. Gas Classification

The BPNN as a classifier has been applied to implement the gas identification function. Figure 2 shows the topology of the BPNN used in this paper. It has three layers with 4 inputs, 3 outputs, and 10 hidden neurons. The four inputs are the four sensor responses. The three outputs represent the possibility of being identified as each target gas, and the gas with the highest probability will be judged as the identification result.

Before training the network, we need to randomly initialize the weights and biases, take a random real number for each weight and bias, and then start forward pass. First, we can compute a_j^l used to activate the jth neuron in the lth layer [14]:

$$a_j^l = \sigma\left(\sum_k w_{j,k}^l a_k^{l-1} + b_j^l\right) \tag{1}$$

$$\text{Activation function: } \sigma(x) = \frac{1}{1+e^{-x}} \tag{2}$$

where $w_{j,k}^l$ is the weight for the connection from the kth neuron in the $(l-1)^{th}$ layer to the jth neuron in the lth layer; b_j^l is the bias of the jth neuron in the lth layer. Then use a^L (the activation vector output from the network) and y (the expected output) to calculate the cost:

$$\text{Cost function: } C = \frac{1}{2}\sum_j \left(y_j - a_j^L\right)^2 \tag{3}$$

Finally, using the gradient descent method, backpropagation changes the weight w and bias b in the network to change the value of the cost function:

$$\text{Weight update: } w_{j,k}^l(m+1) = w_{j,k}^l(m) - \alpha\frac{\partial C}{\partial w_{j,k}^l} \tag{4}$$

$$\text{Bias update: } b_j^l(m+1) = b_j^l(m) - \alpha\frac{\partial C}{\partial b_j^l} \tag{5}$$

where α is the learning rate.

For each sample, if the error is less than the set threshold or the number of iterations has been reached, the training is terminated. Otherwise, it continues to repeat the forward and back propagation processes.

B. Gas Concentration Prediction

In order to find an algorithm model with good stability and high accuracy, we have used the same training set to train the four models of BPNN, RBFNN, SVM, and hybrid LDA-SVM, which uses LDA to achieve dimensionality reduction for input, and uses SVM model for regression.

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For the concentration prediction of these three target gases, the structure of BPNN used is similar to that used for gas classification. It has three layers with 4 input neurons, 1 output neuron, and 10 hidden neurons, but the learning rate has been adjusted. For the RBF network, because it is a fixed three-layer network structure, there is no need to set the number of hidden

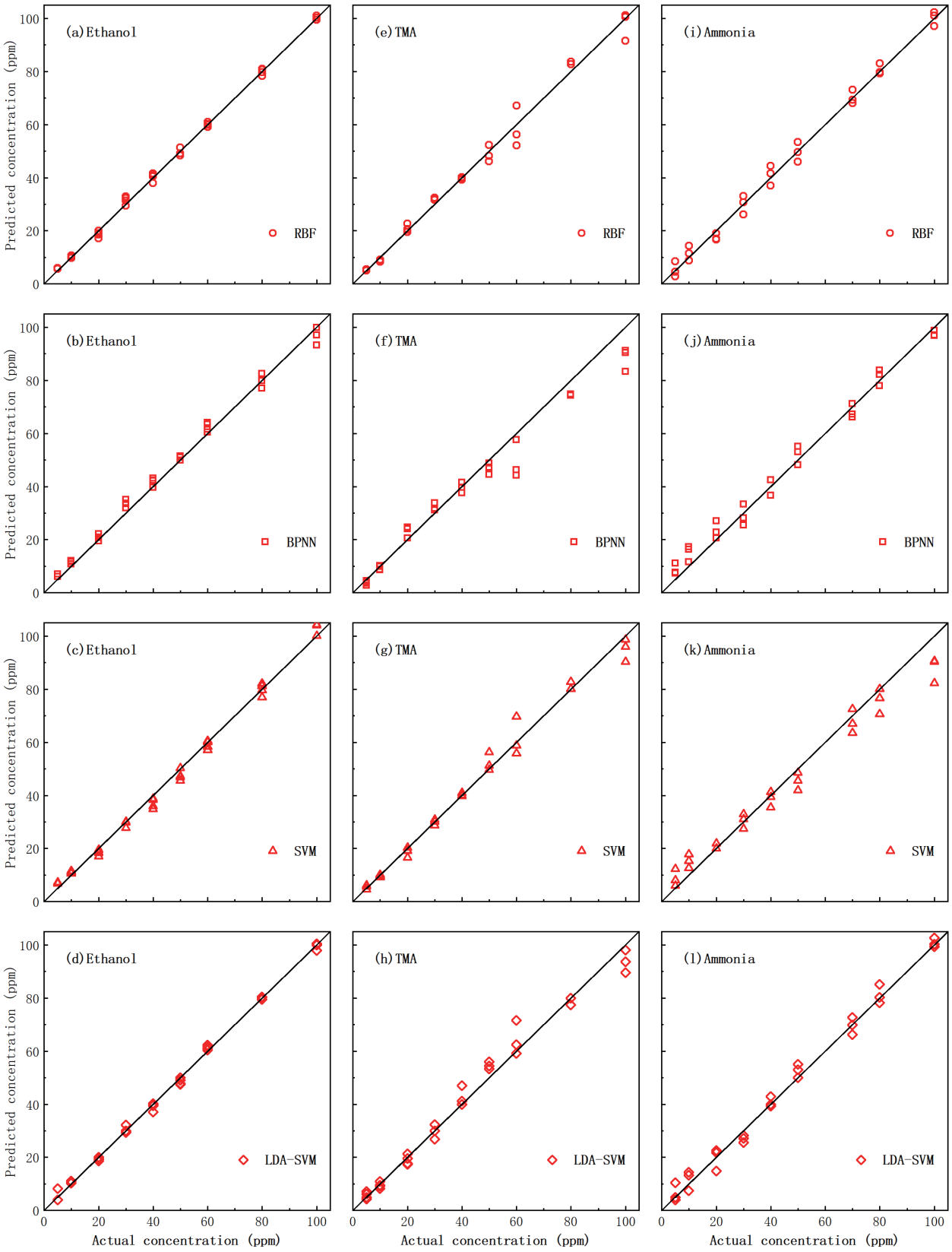


Figure 3. Actual versus predicted concentrations of (a)-(d) ethanol, (e)-(h) TMA and (i)-(l) ammonia using RBF network, BPNN, SVM and LDA-SVM.

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layers. According to the results of multiple simulations, the number of neurons in the hidden layer of the RBFNN concentration prediction model for ethanol, TMA and ammonia is 24, 8, 20 respectively. SVM can perform non-linear classification by kernel method, which is one of the common kernel learning methods [15]. In order to predict the concentrations of the three target gases, based on the simulation results, the SVM model uses a polynomial kernel function, while the hybrid LDA-SVM model chooses a linear kernel function.

III. Results and Discussions

A. Gas Classification Results

In this work, BPNN is used to achieve the function of gas classification. A total of 162 sensor responses were obtained for one sensor for the three target gases in the concentration range of 5-100 ppm.

Therefore, in the gas classification data processing, a 162 × 5 matrix is used (5 represents the responses of the 4 sensors and the label number of this gas), where 75% of the data (121 × 5) is randomly taken as the training set, and the remaining 25% of the data (41 × 5) is used as a test set, and 10 simulations were performed to take the average value. The BPNN model has a 98.8% accuracy rate for gas classification.

B. Gas Concentration Prediction Results

After classifying the gas type through BP neural network, the subsequent step is to import the responses into the corresponding gas concentration prediction model based on the recognition results to estimate its concentration. The prediction results are shown in Figure 3. It can be clearly seen that compared with the other two algorithms, for the concentration estimation of ethanol and ammonia, the prediction results of RBFNN and LDA-SVM are located closer to the actual concentrations, and the prediction results at the same concentration are more stable. However, for trimethylamine, although the correlation between the prediction results of RBFNN and the actual concentration is still superior, the performance of LDA-SVM is slightly worse than SVM. Unlike ethanol and ammonia, the LDA algorithm fails in improving the performance of the SVM model for TMA concentration prediction.

Figure 4 represents a comparison of the relative prediction errors between the four models for the three test gases. It can be seen that at low concentrations, the relative error is generally high; when the actual concentration is greater than 30 ppm, the relative error is reduced. This is because when the concentration is low, the responses of these sensors are weak, and less liquid sample is drawn, so the experimental operation is more demanding and more susceptible to environmental influences. It can also be seen that the prediction of RBFNN at low concentration is more stable than the other three algorithms. Even though the relative error at low concentration is still higher than that at high concentration, the error distribution of RBFNN is more uniform for the concentration range of 5-100 ppm.

In order to compare the four algorithms, four statistical indicators: mean absolute error (MAE), root mean squared error (RMSE), average relative error (ARE), and coefficient of determination (R^2) were used to evaluate the performance of each model, and recorded in Table 1. These parameters have

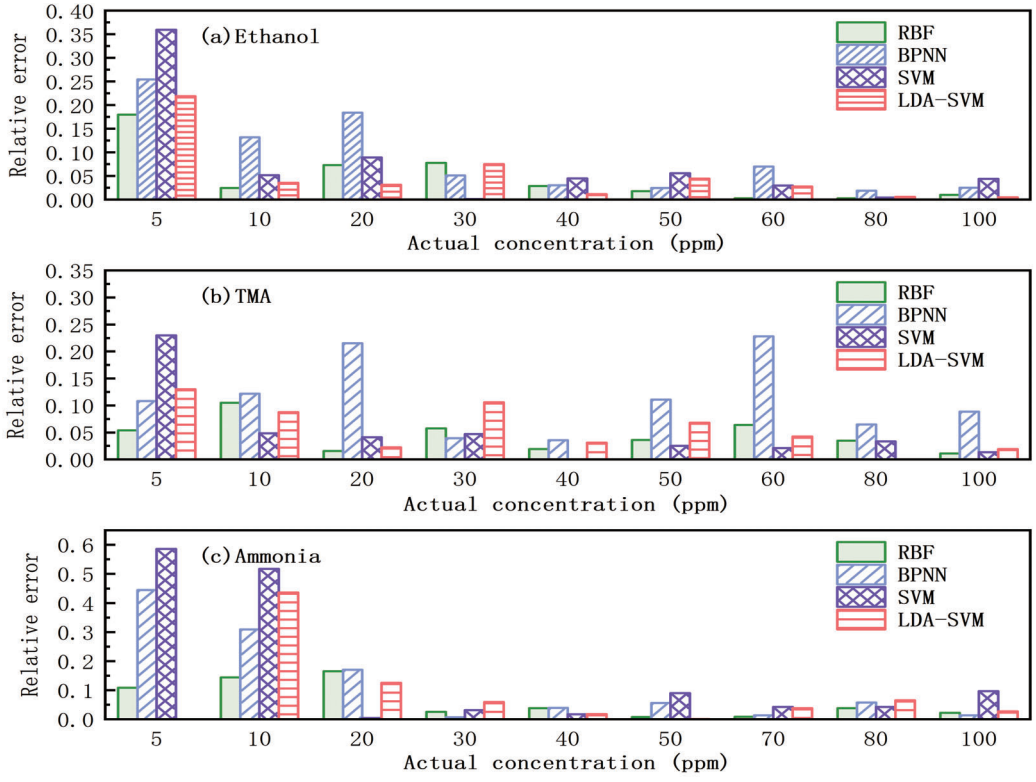


Figure 4. Comparison of the relative errors between these four models for the prediction results of ethanol, TMA and ammonia.

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Table I. COMPARISON OF LDA-SVM, SVM, RBF NETWORK AND BPNN MODELS BY STATISTICS ESTIMATORS.

Algorithm	Ethanol				TMA				Ammonia			
	MAE	RMSE	R^2	ARE	MAE	RMSE	R^2	ARE	MAE	RMSE	R^2	ARE
RBF	0.93	1.14	0.9986	4.64%	1.52	1.88	0.9961	4.41%	1.54	1.86	0.9964	6.21%
BPNN	1.77	2.27	0.9943	8.37%	1.69	2.54	0.9929	6.91%	2.85	3.28	0.9887	12.88%
SVM	1.69	2.11	0.9951	7.55%	1.15	1.34	0.9980	5.11%	3.38	4.36	0.9801	15.87%
LDA-SVM	1.07	1.30	0.9981	5.06%	1.59	1.97	0.9958	5.63%	2.22	2.79	0.9918	8.55%

been defined as below:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_i - \hat{Y}_i\right| \tag{6}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2} \tag{7}$$

$$ARE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right| \tag{8}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{\sum_{i=1}^{n}\left(Y_i - \overline{Y_i}\right)^2} \tag{9}$$

where Y_i means the actual concentration, $\hat{Y}_i$ means the predicted concentration and $\overline{Y}_i$ means the average value.

Table 1 shows that for the concentration estimation of ethanol and ammonia, the RBFNN algorithm is significantly better than the three algorithms of BPNN, SVM and LDA-SVM. Even with a small training set and low sensitivity of commercial sensors, RBFNN still has high prediction accuracy, the mean absolute errors of ethanol and ammonia are approximately equal to 0.93 ppm and 1.54 ppm, while the average relative errors for ethanol and ammonia are less than 5% and 6.5% in the concentration range of 5-100 ppm. For TMA, although the mean absolute error (1.52 ppm) of RBFNN is higher than that of SVM (1.15 ppm), the average relative error of 4.41% is better than 5.11% of SVM. In addition, it can be found that the performance of LDA-SVM is significantly better than SVM for the concentration estimation of ethanol and ammonia; for TMA, although the performance indicators of LDA-SVM is slightly inferior to SVM, it still has high accuracy, and the average relative error is less than 6%, which indicates that combining LDA with SVM may have the ability to improve the performance of SVM for gas concentration prediction in some cases. This also shows that because the response of the sensor array to different gases has different data characteristics, we cannot simply point out which algorithm must be the best for the prediction of gas concentration. During the training process, we also found that when the sample size is not large enough, the outputs of BPNN are not as stable as the other three algorithms. In a conclusion, based on the results of these three different gases predicted by RBFNN, BPNN, SVM and hybrid LDA-SVM, after a series of comparisons and evaluations, we can consider that RBF neural network model in the gas concentration prediction has certain advantages and reliability.

IV. Conclusion

In this work, we combined a gas sensor array consisting of four commercial metal oxide sensors and machine learning algorithms to achieve the functions of gas identification and concentration prediction, thereby further exploring application in monitoring freshness of food such as meat and fruits.

A three-layer BPNN as a classifier has been used to achieve gas recognition, and the average classification accuracy for ethanol, TMA and ammonia can reach 98.8%. For the prediction of the concentration of single gases, we evaluated four algorithms: BPNN, RBF network, SVM, and hybrid LDA-SVM. By analyzing and comparing the prediction results of the four algorithm models, we found that using RBF neural network to estimate the ethanol concentration can achieve peak performance with a mean absolute error of less than 1 ppm and an average relative error of less than 5%. The ammonia concentration prediction with RBFNN achieves a mean absolute error of 1.54 ppm and an average relative error of less than 6.5%, which is significantly better than the results obtained using the other three algorithms. For the concentration estimation of TMA, the average relative error of RBFNN is 4.41%, which is better than 5.11% of SVM. Moreover, for the three target gases, the relative error distribution of RBFNN is more uniform in the concentration range of 5-100 ppm. According to the current prediction results, we can consider that the RBF neural network has certain advantages and reliability in gas concentration estimation, and can basically meet the accuracy requirements for distinguishing food freshness.

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